

*Non-Calculator***Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.****You have 30 minutes.**_____
Signature**SHOW ALL WORK FOR FULL CREDIT!**

Find the product. Write the product in rectangular form, using exact values. (3 points)

1. $\left[\sqrt{6}(\cos 165^\circ + i \sin 165^\circ) \right] \left[\sqrt{2}(\cos 255^\circ + i \sin 255^\circ) \right]$

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

2. $\frac{-\sqrt{3} + i}{\sqrt{2} - i\sqrt{6}}$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

3. $(-1 - i)^3$

Find all complex number solutions of the equation. Write answers in rectangular form. (3 points)

4. $x^3 + 8i = 0$

The fractal called the **Mandelbrot set**: to determine if a complex number $z = a + bi$ is in this set, repeatedly compute $z, z^2 + z, (z^2 + z)^2 + z, [(z^2 + z)^2 + z]^2 + z, \dots$. The complex number z does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise z is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

5. a) $z = i$	YES / NO	b) $z = 1 + i$	YES / NO

ANSWER KEY – Non-Calculator

SHOW ALL WORK FOR FULL CREDIT!

Left: the work, one step per line, the way it goes on paper. Right: why that line happens.

Find the product. Write the product in rectangular form, using exact values. (3 points)

1. $\left[\sqrt{6}(\cos 165^\circ + i \sin 165^\circ) \right] \left[\sqrt{2}(\cos 255^\circ + i \sin 255^\circ) \right]$

WORK

$$\begin{aligned} & \left[\sqrt{6} \operatorname{cis} 165^\circ \right] \left[\sqrt{2} \operatorname{cis} 255^\circ \right] \\ &= \sqrt{6} \cdot \sqrt{2} \operatorname{cis} (165^\circ + 255^\circ) \\ &= \sqrt{12} \operatorname{cis} 420^\circ \\ &= 2\sqrt{3} \operatorname{cis} 60^\circ \\ &= 2\sqrt{3} (\cos 60^\circ + i \sin 60^\circ) \\ &= 2\sqrt{3} \left(\frac{1}{2} + \frac{\sqrt{3}}{2} i \right) \\ &= \boxed{\sqrt{3} + 3i} \end{aligned}$$

Trap: 165° and 255° aren't unit-circle angles; only the sum is. Add first, then look.

WHY

Rewrite each factor as $r \operatorname{cis} \theta$.

Product theorem: multiply the r 's, add the θ 's.

$$\sqrt{12} = \sqrt{4}\sqrt{3} = 2\sqrt{3}. \quad 420^\circ - 360^\circ = 60^\circ.$$

Write cis out.

Unit circle at 60° .

$$\text{Distribute: } 2\sqrt{3} \cdot \frac{\sqrt{3}}{2} = 3.$$

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

2. $\frac{-\sqrt{3} + i}{\sqrt{2} - i\sqrt{6}}$

WORK

$$\begin{aligned} r_1 &= \sqrt{(-\sqrt{3})^2 + 1^2} = \sqrt{3+1} = 2 \\ \theta_1 &= \tan^{-1} \left(\frac{1}{-\sqrt{3}} \right) = -30^\circ \\ \theta_1 &= -30^\circ + 180^\circ = 150^\circ \\ r_2 &= \sqrt{(\sqrt{2})^2 + (-\sqrt{6})^2} = \sqrt{2+6} = 2\sqrt{2} \\ \theta_2 &= \tan^{-1} \left(\frac{-\sqrt{6}}{\sqrt{2}} \right) = \tan^{-1}(-\sqrt{3}) = -60^\circ \\ \theta_2 &= -60^\circ + 360^\circ = 300^\circ \\ \frac{2 \operatorname{cis} 150^\circ}{2\sqrt{2} \operatorname{cis} 300^\circ} &= \frac{2}{2\sqrt{2}} \operatorname{cis} (150^\circ - 300^\circ) \\ &= \frac{1}{\sqrt{2}} \operatorname{cis} (-150^\circ) \\ &= \frac{\sqrt{2}}{2} \operatorname{cis} 210^\circ \\ &= \frac{\sqrt{2}}{2} \left(-\frac{\sqrt{3}}{2} - \frac{1}{2} i \right) \\ &= \boxed{-\frac{\sqrt{6}}{4} - \frac{\sqrt{2}}{4} i} \end{aligned}$$

WHY

Top first: $r = \sqrt{x^2 + y^2}$.

$\tan^{-1}$ only gives QI or QIV.

$x < 0, y > 0$ is QII: add 180° .

Bottom. $\sqrt{8} = 2\sqrt{2}$.

$$\frac{\sqrt{6}}{\sqrt{2}} = \sqrt{3}.$$

$x > 0, y < 0$ is QIV: add 360° .

Quotient theorem: divide the r 's, subtract the θ 's.

$$\text{Rationalize } \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}. \quad -150^\circ + 360^\circ = 210^\circ.$$

210° is QIII: cos and sin both negative.

$$\sqrt{2} \cdot \sqrt{3} = \sqrt{6}.$$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

3. $(-1 - i)^3$

WORK

$$r = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2}$$

$$\theta = \tan^{-1}\left(\frac{-1}{-1}\right) = \tan^{-1}(1) = 45^\circ$$

$$\theta = 45^\circ + 180^\circ = 225^\circ$$

$$(-1 - i)^3 = \left(\sqrt{2} \operatorname{cis} 225^\circ\right)^3$$

$$= (\sqrt{2})^3 \operatorname{cis}(3 \cdot 225^\circ)$$

$$= 2\sqrt{2} \operatorname{cis} 675^\circ$$

$$= 2\sqrt{2} \operatorname{cis} 315^\circ$$

$$= 2\sqrt{2} \left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i\right)$$

$$= \boxed{2 - 2i}$$

WHY

Convert first.

Trap: both parts negative is QIII, not 45° .

De Moivre: r^n and $n\theta$.

$$(\sqrt{2})^3 = \sqrt{2} \cdot \sqrt{2} \cdot \sqrt{2} = 2\sqrt{2}.$$

$$675^\circ - 360^\circ = 315^\circ.$$

315° is QIV: $\cos +$, $\sin -$.

$$2\sqrt{2} \cdot \frac{\sqrt{2}}{2} = 2.$$

Find all complex number solutions of the equation. Write answers in rectangular form. (3 points)

4. $x^3 + 8i = 0$

WORK

$$x^3 = -8i$$

$$r = \sqrt{0^2 + (-8)^2} = 8$$

$$\theta = 270^\circ$$

$$x = \sqrt[3]{8} \operatorname{cis} \left(\frac{270^\circ + 360^\circ k}{3}\right)$$

$$= 2 \operatorname{cis}(90^\circ + 120^\circ k)$$

$$k = 0: 2 \operatorname{cis} 90^\circ = 2(0 + 1i) = 2i$$

$$k = 1: 2 \operatorname{cis} 210^\circ = 2\left(-\frac{\sqrt{3}}{2} - \frac{1}{2}i\right) = -\sqrt{3} - i$$

$$k = 2: 2 \operatorname{cis} 330^\circ = 2\left(\frac{\sqrt{3}}{2} - \frac{1}{2}i\right) = \sqrt{3} - i$$

$$\boxed{x = 2i, -\sqrt{3} - i, \sqrt{3} - i}$$

Trap: $-8i$ is 270° , not 90° . Three roots for a cube, then stop.

WHY

Isolate x^3 : $+8i$ crosses over as $-8i$.

Convert $0 - 8i$.

Straight down the imaginary axis ($x = 0$, so no $\tan^{-1}$).

n th root theorem, $k = 0, 1, 2$.

$$\sqrt[3]{8} = 2. \text{ Cube roots sit } 120^\circ \text{ apart.}$$

QIII: both negative.

QIV: $\cos +$, $\sin -$.

$$\text{Check: } (2i)^3 = 8i^3 = -8i.$$

The fractal called the **Mandelbrot set**: to determine if a complex number $z = a + bi$ is in this set, repeatedly compute z , $z^2 + z$, $(z^2 + z)^2 + z$, $[(z^2 + z)^2 + z]^2 + z, \dots$. The complex number z does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise z is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

5. a) $z = i$	YES / NO	b) $z = 1 + i$	YES / NO
WORK a) $z = i$: $ i = 1$ $z^2 + z = i^2 + i = -1 + i$ $ -1 + i = \sqrt{1+1} = \sqrt{2} \approx 1.4$ $(-1 + i)^2 + i = (1 - 2i + i^2) + i = -i$ $ -i = 1$ $(-i)^2 + i = -1 + i$ YES		WHY $ a + bi = \sqrt{a^2 + b^2}$. Not over 2. $i^2 = -1$. Not over 2, keep going. Square the last result, add z . It repeats $(-1 + i, -i, \dots)$ and never passes 2. Not over 2. <i>Trap:</i> $(1 + i)^2 = 2i$, not $1 + i^2$. Over 2: stop.	
b) $z = 1 + i$: $ 1 + i = \sqrt{2} \approx 1.4$ $z^2 + z = (1 + i)^2 + (1 + i)$ $= (1 + 2i + i^2) + 1 + i = 1 + 3i$ $ 1 + 3i = \sqrt{1+9} = \sqrt{10} \approx 3.2$ NO			